

- 1** A particle is projected with speed u at an angle α above the horizontal from a point O on a horizontal plane. The particle moves freely under gravity.

- (a)** Write down the horizontal and vertical components of the velocity of the particle at time T after projection. [2]

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At time T after projection, the direction of motion of the particle is perpendicular to the direction of projection.

- (b)** Express T in terms of u , g and α . [2]

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- (c)** Deduce that $T > \frac{u}{g}$. [1]

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[illegible]

- (b) Find, in terms of m and g , the tension in the string immediately after P is initially projected horizontally. [4]

[illegible]